MERIDIAN / W104

104-KEY ANSI / BLUETOOTH LE + USB-C / COMPLETE CIRCUIT

KEY MATRIX — six rows / twenty-two columns / COL2ROW

ESC F1 F2 F3 F4 F5 F6 F7 F8 F9 F10 F11 F12 PRINT SCREEN SCROLL LOCK PAUSE

SW1	SW2	SW3	SW4	SW5	SW6	SW7	SW8	SW9	SW10	SW11	SW12	SW13	SW14	SW15	SW16						
COL0	COL1	COL2	COL3	COL4	COL5	COL6	COL7	COL8	COL9	COL10	COL11	COL12	COL15	COL16	COL17						
KEY001	KEY002	KEY003	KEY004	KEY005	KEY006	KEY007	KEY008	KEY009	KEY010	KEY011	KEY012	KEY013	KEY014	KEY015	KEY016						
1u	1u	1u	1u	1u	1u	1u	1u	1u	1u	1u	1u	1u	1u	1u	1u						
D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13	D14	D15	D16						
ROW0	ROW0	ROW0	ROW0	ROW0	ROW0	ROW0	ROW0	ROW0	ROW0	ROW0	ROW0	ROW0	ROW0	ROW0	ROW0						
KEY001	KEY002	KEY003	KEY004	KEY005	KEY006	KEY007	KEY008	KEY009	KEY010	KEY011	KEY012	KEY013	KEY014	KEY015	KEY016						
1N4148W	1N4148W	1N4148W	1N4148W	1N4148W	1N4148W	1N4148W	1N4148W	1N4148W	1N4148W	1N4148W	1N4148W	1N4148W	1N4148W	1N4148W	1N4148W						
GRAVE	1	2	3	4	5	6	7	8	9	0	MINUS	EQUALS	BACKSPACE	INSERT	HOME	PAGE UP	KP NUMLOCK	KP FORWARD SLASH	KP ASTERISK	KP MINUS	
SW17	SW18	SW19	SW20	SW21	SW22	SW23	SW24	SW25	SW26	SW27	SW28	SW29	SW30	SW78	SW79	SW80	SW88	SW89	SW90	SW91	
COL0	COL1	COL2	COL3	COL4	COL5	COL6	COL7	COL8	COL9	COL10	COL11	COL12	COL13	COL15	COL16	COL17	COL18	COL19	COL20	COL21	
KEY017	KEY018	KEY019	KEY020	KEY021	KEY022	KEY023	KEY024	KEY025	KEY026	KEY027	KEY028	KEY029	KEY030	KEY078	KEY079	KEY080	KEY088	KEY089	KEY090	KEY091	
1u	1u	1u	1u	1u	1u	1u	1u	1u	1u	1u	1u	1u	2u	1u	1u	1u	1u	1u	1u	1u	
D17	D18	D19	D20	D21	D22	D23	D24	D25	D26	D27	D28	D29	D30	D78	D79	D80	D88	D89	D90	D91	
ROW1	ROW1	ROW1	ROW1	ROW1	ROW1	ROW1	ROW1	ROW1	ROW1	ROW1	ROW1	ROW1	ROW1	ROW1	ROW1	ROW1	ROW1	ROW1	ROW1	ROW1	
KEY017	KEY018	KEY019	KEY020	KEY021	KEY022	KEY023	KEY024	KEY025	KEY026	KEY027	KEY028	KEY029	KEY030	KEY078	KEY079	KEY080	KEY088	KEY089	KEY090	KEY091	
1N4148W	1N4148W	1N4148W	1N4148W	1N4148W	1N4148W	1N4148W	1N4148W	1N4148W	1N4148W	1N4148W	1N4148W	1N4148W	1N4148W	1N4148W	1N4148W	1N4148W	1N4148W	1N4148W	1N4148W	1N4148W	
TAB	Q	W	E	R	T	Y	U	I	O	P	L BRACKET	R BRACKET	BACKSLASH	DELETE	END	PAGE DOWN	KP SEVEN	KP EIGHT	KP NINE	KP PLUS	
SW31	SW32	SW33	SW34	SW35	SW36	SW37	SW38	SW39	SW40	SW41	SW42	SW43	SW44	SW81	SW82	SW83	SW92	SW93	SW94	SW101	
COL0	COL1	COL2	COL3	COL4	COL5	COL6	COL7	COL8	COL9	COL10	COL11	COL12	COL13	COL15	COL16	COL17	COL18	COL19	COL20	COL21	
KEY031	KEY032	KEY033	KEY034	KEY035	KEY036	KEY037	KEY038	KEY039	KEY040	KEY041	KEY042	KEY043	KEY044	KEY081	KEY082	KEY083	KEY092	KEY093	KEY094	KEY101	
1.5u	1u	1u	1u	1u	1u	1u	1u	1u	1u	1u	1u	1u	1.5u	1u	1u	1u	1u	1u	1u	1u	
D31	D32	D33	D34	D35	D36	D37	D38	D39	D40	D41	D42	D43	D44	D81	D82	D83	D92	D93	D94	D101	
ROW2	ROW2	ROW2	ROW2	ROW2	ROW2	ROW2	ROW2	ROW2	ROW2	ROW2	ROW2	ROW2	ROW2	ROW2	ROW2	ROW2	ROW2	ROW2	ROW2	ROW2	
KEY031	KEY032	KEY033	KEY034	KEY035	KEY036	KEY037	KEY038	KEY039	KEY040	KEY041	KEY042	KEY043	KEY044	KEY081	KEY082	KEY083	KEY092	KEY093	KEY094	KEY101	
1N4148W	1N4148W	1N4148W	1N4148W	1N4148W	1N4148W	1N4148W	1N4148W	1N4148W	1N4148W	1N4148W	1N4148W	1N4148W	1N4148W	1N4148W	1N4148W	1N4148W	1N4148W	1N4148W	1N4148W	1N4148W	
CAPS LOCK	A	S	D	F	G	H	J	K	L	SEMICOLON	QUOTE	ENTER					KP FOUR	KP FIVE	KP SIX		
SW45	SW46	SW47	SW48	SW49	SW50	SW51	SW52	SW53	SW54	SW55	SW56	SW57					SW95	SW96	SW97		
COL0	COL1	COL2	COL3	COL4	COL5	COL6	COL7	COL8	COL9	COL10	COL11	COL12					COL18	COL19	COL20		
KEY045	KEY046	KEY047	KEY048	KEY049	KEY050	KEY051	KEY052	KEY053	KEY054	KEY055	KEY056	KEY057					KEY095	KEY096	KEY097		
1.75u	1u	1u	1u	1u	1u	1u	1u	1u	1u	1u	1u	2.25u					1u	1u	1u		
D45	D46	D47	D48	D49	D50	D51	D52	D53	D54	D55	D56	D57					D95	D96	D97		
ROW3	ROW3	ROW3	ROW3	ROW3	ROW3	ROW3	ROW3	ROW3	ROW3	ROW3	ROW3	ROW3					ROW3	ROW3	ROW3		
KEY045	KEY046	KEY047	KEY048	KEY049	KEY050	KEY051	KEY052	KEY053	KEY054	KEY055	KEY056	KEY057					KEY095	KEY096	KEY097		
1N4148W	1N4148W	1N4148W	1N4148W	1N4148W	1N4148W	1N4148W	1N4148W	1N4148W	1N4148W	1N4148W	1N4148W	1N4148W					1N4148W	1N4148W	1N4148W		
L SHIFT	Z	X	C	V	B	N	M	COMMA	PERIOD	FORWARD SLASH	R SHIFT					UP ARROW	KP ONE	KP TWO	KP THREE	KP ENTER	
SW58	SW59	SW60	SW61	SW62	SW63	SW64	SW65	SW66	SW67	SW68	SW69					SW84	SW98	SW99	SW100	SW102	
COL0	COL1	COL2	COL3	COL4	COL5	COL6	COL7	COL8	COL9	COL10	COL11					COL16	COL18	COL19	COL20	COL21	
KEY058	KEY059	KEY060	KEY061	KEY062	KEY063	KEY064	KEY065	KEY066	KEY067	KEY068	KEY069					KEY084	KEY098	KEY099	KEY100	KEY102	
2.25u	1u	1u	1u	1u	1u	1u	1u	1u	1u	1u	2.75u					1u	1u	1u	1u	1u	
D58	D59	D60	D61	D62	D63	D64	D65	D66	D67	D68	D69					D84	D98	D99	D100	D102	
ROW4	ROW4	ROW4	ROW4	ROW4	ROW4	ROW4	ROW4	ROW4	ROW4	ROW4	ROW4					ROW4	ROW4	ROW4	ROW4	ROW4	
KEY058	KEY059	KEY060	KEY061	KEY062	KEY063	KEY064	KEY065	KEY066	KEY067	KEY068	KEY069					KEY084	KEY098	KEY099	KEY100	KEY102	
1N4148W	1N4148W	1N4148W	1N4148W	1N4148W	1N4148W	1N4148W	1N4148W	1N4148W	1N4148W	1N4148W	1N4148W					1N4148W	1N4148W	1N4148W	1N4148W	1N4148W	
L CONTROL	L GUI	L ALT	SPACE	R ALT	R GUI	APPLICATION	R CONTROL									L ARROW	DOWN ARROW	R ARROW	KP ZERO	KP PERIOD	
SW70	SW71	SW72	SW73	SW74	SW75	SW76	SW77									SW85	SW86	SW87	SW103	SW104	
COL0	COL1	COL2	COL3	COL4	COL5	COL6	COL7									COL15	COL16	COL17	COL18	COL20	
KEY070	KEY071	KEY072	KEY073	KEY074	KEY075	KEY076	KEY077									KEY085	KEY086	KEY087	KEY103	KEY104	
1.25u	1.25u	1.25u	6.25u	1.25u	1.25u	1.25u	1.25u									1u	1u	1u	2u	1u	
D70	D71	D72	D73	D74	D75	D76	D77									D85	D86	D87	D103	D104	
ROW5	ROW5	ROW5	ROW5	ROW5	ROW5	ROW5	ROW5									ROW5	ROW5	ROW5	ROW5	ROW5	
KEY070	KEY071	KEY072	KEY073	KEY074	KEY075	KEY076	KEY077									KEY085	KEY086	KEY087	KEY103	KEY104	
1N4148W	1N4148W	1N4148W	1N4148W	1N4148W	1N4148W	1N4148W	1N4148W									1N4148W	1N4148W	1N4148W	1N4148W	1N4148W	

COLUMN BIAS / 22k

R1	R2	R3	R4	R5	R6	R7	R8	R9	R10	R11	R12	R13	R14	R15	R16	R17	R18	R19	R20	R21	R22
+3V3	+3V3	+3V3	+3V3	+3V3	+3V3	+3V3	+3V3	+3V3	+3V3	+3V3	+3V3	+3V3	+3V3	+3V3	+3V3	+3V3	+3V3	+3V3	+3V3	+3V3	+3V3
COL0	COL1	COL2	COL3	COL4	COL5	COL6	COL7	COL8	COL9	COL10	COL11	COL12	COL13	COL14	COL15	COL16	COL17	COL18	COL19	COL20	COL21
22k	22k	22k	22k	22k	22k	22k	22k	22k	22k	22k	22k	22k	22k	22k	22k	22k	22k	22k	22k	22k	22k

RADIO / USB / POWER

U1 — XIAO nRF52840

ROW_DATA	1	D0	
ROW_CLK	2	D1	
ROW_LATCH	3	D2	
COL_DATA	4	D3	
COL_CLK	5	D4	
COL_LOAD	6	D5	
LED_CAPS	7	D6	
		VBUS	14
	13	GND	
	12	3V3	+3V3
	11	D10	ROW_OE
	10	D9	MODE_KEY
	9	D8	LED_SCROLL
	8	D7	LED_NUM
	15	BAT+_lead	VBAT_SW
	16	BAT-_lead	GND

15/16: wire to underside BAT pads.

C1 — 10u / 10V

SERIAL SCAN ENGINE

U2 — SN74HC595D

ROW_DATA	14	SER	
ROW_CLK	11	SRCLK	
ROW_LATCH	12	RCLK	
ROW_OE	13	OE	
+3V3	10	SRCLR	
+3V3	16	VCC	
GND	8	GND	
		QA	15
	1	QB	ROW1
	2	QC	ROW2
	3	QD	ROW3
	4	QE	ROW4
	5	QF	ROW5
	6	QG	
	7	QH	
	9	QH_serial	

U3 — SN74HC165D

COL0	11	A	
COL1	12	B	
COL2	13	C	
COL3	14	D	
COL4	3	E	
COL5	4	F	
COL6	5	G	
COL7	6	H	
		QH	9
		QH	7
		SER	10
	2	CLK	CHAIN1
	1	SH/LD	COL_CLK
	15	CLK_INH	COL_LOAD
	16	VCC	GND
	8	GND	+3V3

U4 — SN74HC165D

COL8	11	A	
COL9	12	B	
COL10	13	C	
COL11	14	D	
COL12	3	E	
COL13	4	F	
COL14	5	G	
COL15	6	H	
		QH	9
		QH	7
		SER	10
	2	CLK	CHAIN2
	1	SH/LD	COL_CLK
	15	CLK_INH	COL_LOAD
	16	VCC	GND
	8	GND	+3V3

U5 — SN74HC165D

COL16	11	A	
COL17	12	B	
COL18	13	C	
COL19	14	D	
COL20	3	E	
COL21	4	F	
+3V3	5	G	
+3V3	6	H	
		QH	9
		QH	7
		SER	10
	2	CLK	CHAIN2
	1	SH/LD	COL_CLK
	15	CLK_INH	COL_LOAD
	16	VCC	GND
	8	GND	+3V3

C2 — 100n

C3 — 100n

C4 — 100n

C5 — 100n

#FLG01 — PWR_FLAG

#FLG02 — PWR_FLAG

#FLG03 — PWR_FLAG

R26 — 100k / OE off

BATTERY / STATUS

J1 — Protected 1S LiPo

SW105 — BATTERY ON / OFF

SW106 — USB / BLE toggle

R23	LED_CAPS	LED_A0	1k
R24	LED_NUM	LED_A1	1k
R25	LED_SCROLL	LED_A2	1k

LED1 — CAPS

LED2 — NUM

LED3 — SCROLL

OFF isolates cell; USB still powers.

MOUNTING / 1.6 mm PCB / FOUR COPPER LAYERS

H1	H2	H3	H4	H5	H6	H7
M3	M3	M3	M3	M3	M3	M3
5 / 5 mm	229.075 / 5 mm	453.15 / 5 mm	5 / 159 mm	229.075 / 159 mm	453.15 / 159 mm	304 / 110 mm

F.Cu signal / In1.Cu signal / In2.Cu ground / B.Cu signal. PCB 458.15 x 164.00 mm.

Engineering prototype — physical verification required
W104 | 104 keys | BLE + USB | 3.3 V logic
MERIDIAN / original hardware

Sheet: /
File: Meridian W104.kicad_sch

Title: Meridian W104 — complete wireless keyboard

Size: A2	Date: 2026-10-04	**Rev: 2.0**
KiCad E.D.A. 10.0.5		Id: 1/1